

Introduction to Professional Practices in IT

CS4001 – Professional Practices in IT

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Management Accounting

Introduction to Management Accounting

- Focuses on **internal decision-making**, unlike financial accounting.
- Provides **real-time, actionable data** for managers.
- Aids in **planning, monitoring, and control** of business operations.

Example: A company tracks current sales trends to decide whether to ramp up production.

Difference Between Financial & Management Accounting

- **Financial Accounting:** Historical, external use, standardized reports.
- **Management Accounting:** Current/future-oriented, internal use, flexible format.

Example:

- Financial: Annual report for shareholders.
- Management: Weekly sales report for regional managers.

Importance of Planning & Information

- Helps in **setting goals and benchmarks.**
- Essential for **monitoring actual performance vs. budget.**
- Supports **informed decision-making.**

Example: Comparing monthly actual expenses to the budget to identify overspending early.

Key Documents in Management Accounting

- **Budget** – Planned revenues and expenses.
- **Cash Flow Forecast** – Expected inflows and outflows.
- **Variance Analysis** – Comparison between planned and actual figures.

Need for Up-to-Date Financial Information

- Timely data improves **reactive and proactive decision-making**.
- Enables **course correction** before issues escalate.

Example: Immediate sales drop alerts marketing to launch a promotion.

BUDGETS

- A **budget** is a **financial plan** outlining expected income and expenses.
- Typically made for a **fixed period** (e.g., 1 year).
- Helps businesses:
 - Plan resources
 - Monitor performance
 - Make strategic decisions

OVERHEADS

- **Overheads** are the **indirect costs** of running a business that **cannot be directly traced** to a specific product or service. These are essential for operations but are not part of the direct production process.
- They are **recurring expenses** required to keep the business functioning, even if no products are being made or sold.

OVERHEADS

Example from the Computer Assembly Company:

Assume the company has the following overheads:

- **Office rent:** £1,000/month
- **Admin assistant's salary:** £600/month
- **Van fuel and maintenance:** £300/month
- **Electricity:** £100/month

These costs are **not linked to a single computer model** but are needed to run the business.

Table 7.1 Direct costs and expected sales

Model	Cost of components	Technician time	Expected sales
Basic	£200	10 hours	250
Advanced	£300	12 hours	100
Professional	£400	15 hours	50

Table 7.2 An example budget

OVERHEAD EXPENDITURE	
Owner's payroll costs	42,000
Admin assistant's payroll costs (part time)	8,000
Costs of van (including depreciation)	3,500
Internet connection, telephone, postage, etc.	1,000
Advertising	2,000
Premises (heating and lighting, rent, rates, etc)	4,500
Professional fees	1,000
Insurance	500
<i>Total overheads</i>	62,500
OPERATING COSTS	
Technicians' payroll costs	101,160
Bought-in components	100,000
<i>Total operating costs</i>	201,160
Total costs	263,660
SALES INCOME	
Basic model (250 @ £595)	148,750
Advanced model (100 @ £795)	79,500
Professional model (50 @ £895)	44,750
<i>Total income</i>	273,000
Profit	9,340

COST OF LABOUR

- The cost of employing someone is more than just the cost of their salary.
- In most countries, employers are required to pay a tax for every employee. This tax usually goes by a name such as *employers' national insurance contribution* (in the UK) or *social security contribution*; it is proportional to the employee's salary.
- In some countries, this contribution may be as large as 60 per cent of salary, while in others it is very much smaller.

Scenario

Consider one of the technicians who is employed to assemble the computers and suppose that they are paid an annual salary of £30,000. To calculate the cost of the time the technician spends assembling a computer, the annual payroll cost is not very helpful. What we need to know is the cost per hour. This is harder to calculate than we might expect. First, we need to calculate how many days we can expect the technician to work. There are 52 weeks in a year. Assuming the company works a five-day week, there will be 260 weekdays. However, the company will be closed for public holidays and the technician will not be working. The number of public holidays varies from country to country and even within a country.

It tells how all these factors can be taken into account to obtain a figure for the number of revenue-earning working hours that can be expected from an employee over the period of one year.

Table 7.3 Calculation of the number of revenue-earning hours in a year

Total number of weekdays (1)	260
Public holidays (2)	10
Annual leave (3)	28
Sick leave (4)	5
Unproductive time (5)	15

Total non-revenue-earning time (6)	58
<i>Calculated as: (2) + (3) + (4) + (5)</i>	
Total number of revenue-earning days (7)	202
<i>Calculated as: (1) – (6)</i>	
Total number of hours available	1,414
<i>Calculated as: (7) × 7</i>	

Scenario

In the UK, the law requires employers to make a contribution to a pension scheme. The details are complicated but we shall assume here that the employer's contribution is equal to 3% of an employee's salary. The employer also has to pay what is called the Employer's National Insurance Contribution.

Table 7.4 Calculation of the hourly cost of labour

Annual salary		£30,000
Employer's pension contribution	3% of £30,000	£900
Employer's National Insurance Contribution	13.8% of £(30,000–9,564)	£2,820
<i>Total annual cost</i>		£33,720
<i>Hourly direct cost</i>	£33,720 / 1,414	£23.85

The direct annual cost of employing the technician will thus be £30,000 plus the 3% pension contribution and the Employer's National Insurance Contribution, which is 13.8% of (£30,000 – £9,564). This leads to an annual direct cost of £33,720, so that the direct cost of an hour of a technician's time is $£33,720 / 1,414 = £23.85$.

Scenario Cont.

In a large business, there are many ways in which overheads can be spread over the different activities the business performs; there is no single 'right' way of doing this. One way or another, the company has to recover the overhead costs of £62,500. The only way it can do this is collect it from the income it gets from sales. However, there are many different ways in which this can be done. We shall describe three of the most common ways of doing this. The simplest way is to allocate the same fixed overhead to each computer sold, regardless of the cost of the components or the amount of labour involved. Since we expect to sell 400 units, this means £156.25 per computer.

Way 1: Same fixed overhead to each item

Cost per computer + Cost of components + (Technician time x direct cost of an hour of a technician's time)

The Basic model would then cost
 $£156.25 + £200 + 10 \times £23.85 = £594.75$

The Advanced model would cost
 $£156.25 + £300 + 12 \times £23.85 = £742.45$

The Professional model would cost
 $£156.25 + £400 + 15 \times £23.85 = £914.00$

Way 2: Proportional to the number of hours of labour involved.

The second way of allocating the overhead is to make it proportional to the number of hours of labour involved. This means adding an overhead component to the cost of an hour of a technician's time. Since we have three technicians, each supplying 1,414 hours of productive labour per year, we need to add

$$£62,500 / (3 \times 1,414) = £14.73$$

the cost of an hour's labour, making it up to $£23.85 + £14.73 = £38.58$.

Now the cost of the three models comes out at

$$£200 + 10 \times £38.58 = £585.80 \text{ (Basic)}$$

$$£300 + 12 \times £38.58 = £762.96 \text{ (Advanced)}$$

$$£400 + 15 \times £38.58 = £978.70 \text{ (Professional)}$$

Way 3: Distribute the overhead in proportion to the total cost

We want to **distribute overhead costs (£62,500)** fairly across three different product models: **Basic, Advanced, and Professional.**

Step-by-Step Explanation:

1. Calculate direct costs for each model

This includes:

- **Component cost** (a fixed amount)
- **Labour cost** (hourly rate × number of hours)
- The hourly rate is **£23.85**.

Way 3 Cont.

Model	Component	Hours	Labour ($23.85 \times$ hours)	Total Direct Cost
Basic	£200	10	£238.50	£438.50
Advanced	£300	12	£286.20	£586.20
Professional	£400	15	£357.75	£757.75

Way 3 Cont.

2. Multiply each model's cost by how many units will be sold

We're expecting to sell:

250 Basic units - 100 Advanced units -50 Professional units

So the total direct cost is:

$$250 \times 438.50 = \text{£}109,625.00$$

$$100 \times 586.20 = \text{£}58,620.00$$

$$50 \times 757.75 = \text{£}37,887.50$$

$$\text{Total} = \text{£}206,132.50$$

Way 3 Cont.

3. Spread the £62,500 overhead across all products

To do this, we find out **what % of the direct cost this overhead is:**

$$62,500 \div 206,132.50 = 0.30 \text{ (or 30\%)}$$

So we increase each product's cost by **30%** to include overhead.

Way 3 cont.

Model	Direct Cost	Overhead (30% of DC)	Final Cost
Basic	£438.50	£131.55	£570.05
Advanced	£586.20	£175.86	£762.06
Professional	£757.75	£227.33	£985.08

Comparisons

	Fixed overhead	Overhead proportional to Labour hour	Overhead proportion to the total cost
Basic	595	586	570
Advanced	742	763	762
Professional	914	979	985

The costs calculated in this way can form the basis for pricing the computers. Certainly, computers should not normally be sold at prices that are lower than the costs. More commonly, the prices will be set on the basis of 'what the market will bear', that is, how much customers will be willing to pay, and this depends very much on the competition. The cost calculations will, however, show us where our costs lie and how to go about reducing them so that we can sell our products more cheaply.